

PACK WITH LOVE CLUB

From Data to Impact · Course Map

What each lesson builds — business, math, statistics — and where it sits in the revised AP Statistics course · 2026–27 · companion to the 24-Lesson Curriculum

Reading the AP column. Topic numbers are Unit.Topic (1.1–1.13; 2.1–2.12). Skills: 1.A determine an investigative question · 2.A identify information · 2.B justify an ethical collection method · 3.A construct representations · 3.B calculate summary statistics · 3.C estimate counts, percentages, probabilities · 4.A describe and compare · 4.B justify a claim · 4.C describe distributions · 4.D interpret results. Practices: P1 Formulate Questions · P2 Collect Data · P3 Analyze Data · P4 Interpret Results. Concept level only; inference (Units 3–5) is out of scope by design.

#	Date	Lesson	Practice	Business capability	Math capability (modeling & quantitative reasoning)	Statistical reasoning	AP (topics · skills)
1	9/13	What Is a Business? (Prof. Sun)	—	Four-box business model; recognizing risk; who decides on funds	Revenue – Cost = Profit as an equation; price × units; estimation (MP4, MP2)	Seed: how would you find out?	seed of 1.A
2	9/20	How Products Move (Tony Yao)	—	Supply-chain literacy; three entry points; MOQ, sampling, defects, lead time	Order of magnitude; cost/time/risk as three dimensions of one decision	—	—
3	9/27	Choose the Product, Find the Source (Joyce Xie)	—	Product screening; reading a listing; listed vs. quoted; landed-cost estimate; inquiry drafting; product decision	First landed-cost model — multi-step percent, unit rate (7.RP); a source and an uncertainty per number (MP6)	Source-reliability habits (P2/P4, not yet named)	habits for 2.A / 4.D
4	10/4	Pose the Statistical Question — Who Are We Asking?	Question	Turning a business unknown into a researchable question	—	Investigative question, three components, fixed; population/sample/census; observational units; variables; parameter vs. statistic named; prediction in context	1.1 (1.1.A, 1.1.B taught) · 1.2 · 1.10.A–B · 1.10.D.4 · 1.A, 2.A
5	10/11	Bias, Two Ways — Whom We Ask and How We Ask	Collect	School-approvable research; School Support Request	Fractions and quotas	SRS, stratified, convenience; bias = systematic error; voluntary response, undercoverage; wording bias; honest generalization; Survey v1	1.11 · 1.12 · 1.10.E · 2.A, 2.B
6	10/18	Survey Launch Lab → Survey Week 10/18–25	Collect	Field discipline; ethical customer contact	Coding scheme	Simulation of friends-only bias; quota cards; pilot; interviewer effect; nonresponse as data	2.3 · 1.12 · 1.10.E · 2.A, 2.B, 3.C
7	10/25	Raw Data → Percentages & Averages	Analyze	Reading a market	Fraction → percent by hand; mean, median, range; five-number summary (ext.)	Distributions; parameter vs. statistic; sampling variability by school; $1/\sqrt{n}$	1.2.A · 1.3 · 1.5–1.9 · 2.12 · 3.A, 3.B, 4.A, 4.C
8	11/1	Graphs, Evidence & the Price Signal	Analyze → Interpret	Market brief; segments; price range; skepticism toward AI	Row percentages; reading a truncated axis	Joint/marginal/conditional; association; CER; predictions opened	1.4 · 1.9 · 2.1–2.2 · 3.A, 3.B, 4.A, 4.B, 4.D
9	11/8	Samples in Hand — the Mathematics of a Weighted Decision	—	Supplier evaluation: sample QC at home; quoted vs. listed; the Jan 10 knock-out; China and US wholesale on equal footing; weighted matrix (price as landed cost); recommendation	Weighted average $\Sigma(w \times s) \div 100$ and why weights total 100; the flip-point inequality (7.EE.4)	Evidence-cited scoring	3.B · 7.EE.4
—	11/15	<i>no class</i>					
10	11/22	Cost per Unit, Quantity & the Order (order ≈ 11/23)	—	Unit economics; quantity with buffer; budget ceiling; go/no-go; confirming the path	Cost model fixed + variable × qty; unit rate (7.RP.1); percent buffer (7.RP.3); budget as inequality (7.EE.4)	Optimism haircut	—
—	11/29	<i>Thanksgiving</i>					
11	12/6	Landed Cost — the Real Numbers, and What We Didn't Buy	—	Import math on the chosen path; domestic vs. import; tariff risk; MOQ trap in hindsight	Multi-step percent with real numbers; percent error; sensitivity one input at a time (MP2)	—	—
12	12/13	Pricing Scenarios — the Data Sets the Price	Interpret (cycle 1 closes)	Pricing strategy; break-even; sell-through	Scenario table; break-even = fixed ÷ (price – unit cost) (MP4)	Discounting stated intent; assumptions named	4.D habit

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—	12/20, 12/27	winter break					
13	1/3	Mission-Based Marketing	Question → Collect (cycle 2 opens)	Value proposition; Charter §8 messaging; message test	—	Experiment: units, treatments, response, random assignment by coin flip; causal scope	1.1.B applied · 1.10.C · 1.13.A, D · 1.A, 2.A, 2.B
14	1/10	The Boxes Are Open — Arrival Inspection, and What a Sample Can Tell You	Collect → Analyze	Receiving; defect definition; acceptable rate; claim path; risk register (business 15 min)	Percent of a quantity; count reconciliation	Random sample of units; simulation of a 3% batch — fifteen draws of 40 into a dotplot ; sampling distribution of a proportion	1.2 · 1.11 · 2.3 · 2.12 · 2.A, 3.A, 3.C, 4.C
15	1/17	Forecast Demand & Choose the Market	Question → Analyze	Demand forecast; one market for both events ; audience-fit gap; adult mini-survey	Multiplicative model traffic × stop × buy; expected revenue = price × buyers	Random process; law of large numbers; expected value; to whom conclusions apply	1.1.B applied · 1.10.E · 2.3 · 2.4 · 2.9 · 1.A, 2.A, 3.C
16	1/24	Sales System, Event Day Kit & Two Ways to Run an Experiment	Collect (design)	Point-of-sale system; cash controls; roles; market outreach	Conversion = buyers ÷ stopped; revenue per buyer; units per hour	Design 1 whole-event (no causal claim) vs. Design 2 randomized alternating slots (causal claim, scoped); confounding; replication	1.10.C–D · 1.13.A, D · 2.A, 2.B
17	1/31	Home Practice Sale — and Turning the Forecast into a Distribution	Collect → Analyze	Selling at home; reconciliation; Kit v2; inventory split (business 20 min)	Units × price = cash; a percentile-based stock decision	Simulation of fifteen Saturdays into a distribution ; expected value with variability	2.3 · 2.9 · 2.12 · 2.A, 3.A, 3.C, 4.C
18	Sat 2/6	Sales Event I (same market)	Collect	Running a real sale; cash rule; adult mini-survey at the table	Real-time tally; running conversion	Observations on defined variables; the baseline	1.2 · 2.A
19	2/7	Analyze & Adjust — Which Variables May Be Randomized?	Analyze → Interpret → Question	One change, committed in writing, with its design	Percent error by input; percent change; numeric prediction	Stated vs. revealed (students and adults); slot-switchable vs. whole-event-only adjustments; price cannot ethically be randomized ; Event II question fixed	1.1.B student-generated · 1.13.A · 2.12 · 1.A, 3.B, 4.B, 4.D
—	2/14	reserve Sunday					
20	Sat 2/20	Sales Event II / Campaign Close (same market)	Collect	Executing the adjustment on the coin-flip slot schedule; unsold accounting	Cumulative totals; comparison by slot	The randomized schedule executed; replication	1.13.A · 2.A
21	2/21	Post-Sale Statistical Analysis	Analyze → Interpret	Campaign review by segment; honest attribution; an inconclusive result reported as a finding	Percent change; ratio stated/revealed; hand-verification (MP6)	A/B two-way table and its causal scope ; whole-day comparison and its limits; confounding; verify A1	2.1–2.2 · 1.13.D · 1.10.D.5 · 3.A, 3.B, 4.A, 4.B, 4.D
22	2/28	Profit, Net Fundraising & Impact Math	—	Net vs. gross; inventory cost; impact accounting; explaining to a parent	Revenue – ordered cost – expenses = net; margin; net as % of \$5,300	Percent of a whole, honestly labeled	—
23	3/7	Business Review	Interpret (cycle 2 closes)	Post-mortem; decision vs. outcome; Year 2 Brief	2×2 reasoning under uncertainty	Evidence synthesis; the question in reverse	1.1.B reflected · 4.B, 4.D
24	Sun 3/14	Final Impact Showcase	All four	Public accountability; portfolio	One verified number per beat	Data storytelling	1.A–4.D · P1–P4

Coverage summary

- **Unit 1 (Exploring One-Variable Data and Collecting Data):** all 13 topics at the concept level — 1.1–1.2 (L4, L7), 1.3–1.9 (L7–L8), 1.10 (L4, L5, L16), 1.11 (L5, L14), 1.12 (L5–L6), 1.13 (L13, L16–L21).
- **Unit 2 (Probability, Random Variables, and Probability Distributions):** 2.1–2.2 (L8, L21), 2.3 (L6, L14, L15, L17), 2.4 (L15), 2.9 (L15, L17), 2.12 by simulation and intuition (L7, L14, L17, L19).
- **Skills:** 1.A, 2.A, 2.B, 3.A, 3.B, 3.C, 4.A, 4.B, 4.C, 4.D — the ten non-inference skills, and the two free-response questions (FRQ 1: Practices 1–2; FRQ 2: Practices 3–4) that assess them.
- **Lesson types:** 9 led by statistics (L4–L8, L14–L17 in their second halves, L19, L21) · 4 by business math (L10–L12, L22) · 7 by business and operations (L1–L3, L9, L13 first half, L14 and L17 first halves) · 2 fieldwork (L18, L20) · 2 synthesis (L23, L24).